Mini Diffusion Model for MNIST: A Comparison of DDPM and DDIM Sampling

DATA 266 -Homework 8 | SID4: 1598

Abstract—This report studies a small diffusion model trained on MNIST handwritten digits. A six-layer convolutional neural network (CNN), conditioned on the diffusion timestep, was trained to estimate Gaussian noise over 50 epochs. Its training mean squared error (MSE) fell from 0.1775 to 0.0602. We then generated ten images each using 100-step DDPM and 50-step and 20-step DDIM sampling. The generated images showed handwritten-like strokes but were generally difficult to recognize. Five-run timing tests showed that 20-step DDIM was faster than 50-step DDIM (0.0428 versus 0.1172 seconds per batch). The report discusses the speed–quality tradeoff and a possible mismatch between terminal training noise and Gaussian sampling initialization.

Index Terms—diffusion models, DDPM, DDIM, MNIST, noise prediction, generative models.

I. INTRODUCTION

Diffusion models generate images by learning to reverse a noise-adding process. In the forward process, random Gaussian noise is mixed with clean images. A neural network is trained to predict that noise, making reverse generation possible. This homework implements the forward diffusion equation, trains a compact CNN, and compares DDPM [1] against the faster-step DDIM method [2] on MNIST [3].

II. METHODOLOGY

The MNIST training set contains 60,000 grayscale 28 × 28 images, normalized to [−1, 1]. The forward process uses T = 100 steps with a linear beta schedule from 0.0001 to 0.02. At a selected step t, $x_t = \sqrt{(\bar{\alpha}_t)}x_0 + \sqrt{(1 - \bar{\alpha}_t)}\varepsilon$, where ε is standard Gaussian noise. The notebook shows example forward-noised digits at t = 0, 10, 30, 60, and 100.

The noise predictor contains six convolutional layers and ReLU activations, together with a learned 32-dimensional timestep embedding. The network predicts a noise tensor with the same dimensions as the noisy input. Training used Adam with learning rate 0.001, batch size 128, and MSE loss over 50 epochs. Reverse generation starts from Gaussian noise. DDPM uses 100 stochastic reverse steps; deterministic DDIM ($\eta = 0$) uses 50 or 20 selected steps.

III. RESULTS

The recorded first-epoch training MSE was 0.1775, and the last-epoch MSE was 0.0602, corresponding to a 66.1% reduction. Most improvement occurred early, followed by slower changes toward the end. This is evidence that noise prediction improved during training, not a direct measure of generated-image realism.

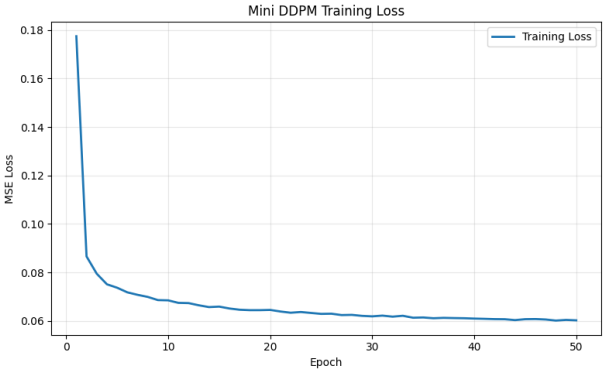


Fig. 1. Training MSE across 50 epochs (notebook output).

IV. SAMPLING COMPARISON

All three methods generated ten image tensors of shape 1 × 28 × 28. In the displayed samples, most digits were unclear, with overlapping or heavy handwritten-like strokes. The DDIM 50-step and 20-step images were visually similar. Their displayed rows shared initial Gaussian noise, but the previously generated DDPM row used a different starting noise; thus the figure is descriptive rather than a paired comparison across all methods.

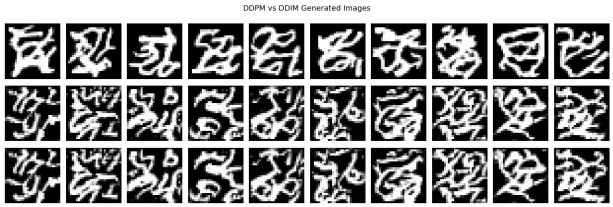


Fig. 2. Generated images: DDPM (100), DDIM (50), and DDIM (20) steps.

After a warm-up, five repeated timing runs gave 0.1172 ± 0.0158 seconds per batch for DDIM-50 and 0.0428 ± 0.0015 seconds for DDIM-20. In this measurement, DDIM-20 was about 2.74 times faster. DDPM sampling time was not measured in the original notebook.

V. FAILURE ANALYSIS AND LIMITATIONS

The network learned to reduce noise-prediction error, but this did not translate into clearly recognizable generated digits. One possible reason is the short schedule: the cumulative alpha at the final timestep is about 0.364, so training examples at the terminal step retain substantial signal from real digits. Sampling instead begins from pure Gaussian noise. This difference may affect generation quality, but the experiment does not isolate it as the sole cause. Limited CNN capacity may also contribute. No FID, generated-digit accuracy, or multiple-seed quality evaluation was collected.

VI. CONCLUSION

This mini implementation completed forward diffusion, CNN noise-prediction training, and DDPM/DDIM image generation. DDIM with 20 steps offered substantially lower measured sampling time than DDIM with 50 steps, without an obvious visual difference in this small set of outputs. Image quality remained weak despite decreasing MSE. A longer diffusion schedule, a stronger noise predictor, and quantitative image evaluation are useful follow-up experiments.

REFERENCES

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